



SCHOOL OF COMPUTER TECHNOLOGY

AASD 4013 Agile methodologies

Deadline:

Presentation date: Last week of class,

Final submission due date: One week after the last class

Project Overview

The goal of this project is to **gain hands-on experience with agile software development**, using tools such as **Jira** or **Azure DevOps** (students may choose the tool they prefer).

You will work on a **code development project**, which can either be:

- A new project defined specifically for this course, or
- A project from another course you're currently enrolled in.

Projects may involve **neural network (NN) model training** or work with **signals, time series, text, or images** you have full freedom in choosing the domain.

Important: The focus of this project is **not on the final results**, but **on how you develop the code using agile methodologies**. However, if you follow an organized, iterative approach, you should still be able to achieve meaningful outcomes.

Project Report Requirements

Your final report should include the following key sections:

1. Problem Definition

Clearly define your project by covering the following:

- **Goal and scope** of the project
- **Success criteria** (how you'll know if your project is successful)
- **Stakeholder and user discussions**
- **Basic project management** elements, including:
 - Scope
 - Budget (if any)
 - Timeline
 - Deliverables
 - Team members/resources
 - Risks and mitigation strategies

2. Agile Methodology & Project Workflow (Scrum)

Demonstrate how you applied Scrum methodology throughout the project.

You must simulate **two complete Scrum cycles (sprints)**, each lasting **one week**, for a total of **2–3 weeks** of development.

Include the following in your report:

A. Team Roles

Define each team role clearly:

- **Scrum Master**
- **Product Owner**
- **Development Team Members**
- **End Users / Stakeholders**

B. Scrum Activities & Artifacts

For **each sprint**, include:

1. **Sprint Planning**
 - Meeting minutes
 - Screenshot of the sprint board from your chosen tool
2. **Daily Scrums**
 - Include notes or summaries for **at least 2 daily stand-ups per sprint**
3. **Sprint Review**
4. **Sprint Retrospective**
5. **Backlogs**
 - **Product Backlog** (features, enhancements, etc.)
 - **Sprint Backlog** (tasks, user stories, bugs if any)
 - Examples of **user stories**, **bugs**, and **tasks**
6. **Scrum/Task Boards**
 - Screenshots from your tool (Jira/Azure DevOps/etc.)
7. **Burndown Charts**
 - Include a snapshot or export of your team's burndown report

Note: Not all elements need to be exhaustive provide **representative examples** to show how things evolved over time. Some tools like Jira have these artifacts which can be obtained after setting up a project.

3. Results & Reflections

- Discuss whether the project was a success based on your defined goals.
- Reflect on your **process and methodology**.
- Highlight lessons learned from applying agile development.

Sprint Task Planning – Sample Ideas

When creating your sprint tasks, be as **specific and detailed** as possible. Focus on deliverables that contribute to the final product and tie each task back to stakeholder needs.

Example tasks:

- Collect data (define source, format, tools)
- Preprocess data (e.g., clean missing values, normalize features)
- Develop function to perform task X (e.g., "normalize time series data")
- Perform data analysis (e.g., statistical summaries, visualizations)
- Build neural network model (define architecture, layers, etc.)
- Train and tune the model (hyperparameter optimization)
- Evaluate and test the model (accuracy, precision, etc.)
- Visualize results (graphs, dashboards, etc.)

Guidelines

Criterion	Poor	Good	Great	Amazing	Perfect
Clarity and Completeness of Problem Definition & Project Planning	States a general project topic. Omits or only briefly mentions project goal. No evidence of stakeholder or user consideration. No project management elements included	States a project goal and scope. Lists at least one success criterion. Mentions at least one stakeholder or user. Includes at least one project management element (e.g., timeline or deliverables)	Clearly defines project goal and scope. Lists measurable success criteria. Summarizes discussions or considerations with stakeholders or users. Includes at least three project management elements (scope, timeline, deliverables, team/resources, risks, or budget)	Clearly defines project goal, scope, and measurable success criteria. Summarizes input from multiple stakeholders or users. Includes all required project management elements: scope, timeline, deliverables, team/resources, risks, and budget (if any). Identifies at least one risk and a mitigation strategy	Provides a detailed project goal, scope, and multiple measurable success criteria. Summarizes input from multiple stakeholders or users, including how their needs shaped the project. Includes all project management elements: scope, budget (if any), timeline, deliverables, team/resources, risks, and mitigation strategies. Connects each project management element to the project's overall success
Application and Documentation of Agile (Scrum) Methodology	Mentions use of Scrum or agile, but provides no evidence of process. No team roles defined. No Scrum artifacts or activities included	Lists team roles (Scrum Master, Product Owner, Development Team). Includes at least one Scrum artifact (e.g., a product backlog or sprint board screenshot). Describes at least one Scrum activity (e.g., sprint planning or daily scrum)	Defines all team roles and their responsibilities. Includes representative examples of Scrum artifacts for both sprints (product backlog, sprint backlog, sprint board screenshot, burndown chart). Documents at least one activity per sprint: sprint planning, daily scrum notes, sprint	Defines all team roles, including end users/stakeholders, and describes their involvement. Includes all required Scrum artifacts for both sprints: product backlog, sprint backlog, sprint board screenshots, burndown charts, and user stories/tasks/bugs. Documents all Scrum activities for both sprints: sprint planning (with meeting minutes), at least two daily scrum	Defines all team roles, including end users/stakeholders, and explains how each contributed to the process. Includes all Scrum artifacts for both sprints, with clear evidence of iteration and adaptation (e.g., updated backlogs, evolving boards, changing burndown charts). Documents all Scrum activities for both sprints, including detailed meeting

			review, and retrospective	summaries per sprint, sprint review, and retrospective. Shows evolution of artifacts and activities over time	minutes, daily scrum summaries, sprint reviews, and retrospectives. Provides examples of how feedback or issues from one sprint informed changes in the next
Depth of Results Analysis & Reflection on Agile Process	States whether the project was completed. No discussion of process or outcomes	States whether project goals were met. Briefly mentions one aspect of the process or outcome	Discusses whether project goals and success criteria were met. Describes at least one challenge or lesson learned from the agile process. Provides a basic reflection on the team's workflow	Analyzes project outcomes in relation to all defined success criteria. Reflects on multiple aspects of the agile process, including what worked and what could be improved. Identifies specific lessons learned and how they would inform future projects	Provides a detailed analysis of project outcomes, referencing all success criteria and stakeholder needs. Reflects deeply on the agile process, including team dynamics, adaptation to change, and the impact of Scrum practices. Identifies multiple lessons learned, with concrete examples, and proposes actionable improvements for future agile projects. Connects reflections to broader principles of agile methodology and software development